

Audit of the source Potapov verdict

non-commuting J -factors still cannot carry the negative square of $\mathfrak{T}_0$

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Abstract

Round 431 closed the first source-Potapov discovery leg with `SOURCE_FACTOR_NOT_FOUND`: the unipotent product of point-mass matrices equals the Cauchy m -function of X , whose Pick kernel is positive definite, while $I - \mathfrak{T}_0^* \mathfrak{T}_0$ has one negative square. The factor class is abelian, so the reviewer contract (DCCXCVII) asked whether that collapse is a strawman of commutativity — whether a genuine Blaschke–Potapov J -factor with the chain-scattering Redheffer star could place the square. This note rebuilds that richer class independently (source $\alpha = x + i|w|$, J -positive generating vectors $u = (2, x)$ that do not commute) and proves a small closure lemma: any finite Redheffer product of source-pure *positive-mass* J -factors stays in $\mathcal{S}_0$. The R431 numbers recompute exactly. The Cauchy– π identity holds on a second $\mathbb{Q}$ -window, where the negative squares again live in $W_Y - S_0$ and not in the Euler Pick. Ausgang **CONFIRMED**: the route is cleanly closed. No RH claim.

Status. Lemma 1 is proved (algebra of J -contractions). Propositions 1–2 are independent machine checks of R431. The R431 artefacts are read-only. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 Question 1: is the factor a strawman?

The R431 elementary factor

$$E(z; x, w) = \begin{pmatrix} 1 & 0 \\ w/(x - z) & 1 \end{pmatrix} \quad (1)$$

commutes, and the 2×2 product is m_X . That product *is* the Redheffer star in the Homma port stacking (Proposition 1(iv)): the scattering form of E has $S_{22} = 0$, the star pivot $1 - S_{22}T_{11}$ is 1, and $E(x, w) \star E(x', w')$ adds the masses. Commutativity is a special case, not a substitute for the star.

The richer positive-mass class used here is the Blaschke–Potapov elementary factor

$$B(z; \alpha, u) = I + (b_\alpha(z) - 1) P_u, \quad b_\alpha(z) = \frac{z - \alpha}{z - \bar{\alpha}}, \quad (2)$$

with $P_u = u(u^* J)/(u^* J u)$, $J = \text{diag}(1, -1)$, and source-pure data $\alpha = x + i|w|$, $u = (2, x)$ (so $u^* J u = 4 - x^2 \geq 3$ on $[-1, 1]$). These factors are J -unitary on $\mathbb{R}$ and do not commute ($\| [B_0, B_1] \|_F = 5.561 \cdot 10^{-2}$ at $z = i$ on the five-atom toy).

Lemma 1 (positive-mass J -products stay in $\mathcal{S}_0$). *Let $J = J^* = J^{-1}$ have signature $(1, 1)$. A 2×2 function holomorphic in $\mathbb{C}_+$ is J -contractive there if $\Theta(z)^* J \Theta(z) \leq J$.*

- (i) *The class is closed under pointwise multiplication: $(\Theta_1 \Theta_2)^* J (\Theta_1 \Theta_2) = \Theta_2^* (\Theta_1^* J \Theta_1) \Theta_2 \leq \Theta_2^* J \Theta_2 \leq J$.*

- (ii) Under Homma conversion, the 2×2 scattering Redheffer star equals the chain product, so (i) is the cascade of two-ports. A cascade of contractive scattering matrices is contractive.
- (iii) The factor (2) with $\text{Im } \alpha > 0$ and $u^*Ju > 0$ is J -inner on $\mathbb{R}$ and holomorphic in $\mathbb{C}_+$, hence J -contractive in $\mathbb{C}_+$ by the J -maximum principle. A unipotent Herglotz atom $E(z; x, w)$ with $w > 0$ is the boundary-mass limit of the same class (Nevanlinna functions of a positive measure).
- (iv) Krein–Langer: $\kappa(\Theta_1 \star \Theta_2) \leq \kappa(\Theta_1) + \kappa(\Theta_2)$. Every source-pure positive-mass factor has $\kappa = 0$, so any finite product has $\kappa = 0$. Its J -kernel Pick matrix is positive semidefinite, independently of grouping and of non-commutativity, and cannot be congruent to a one-defect $I - \mathfrak{T}_0^* \mathfrak{T}_0$.

The inverse-Blaschke b_α^{-1} (pole in $\mathbb{C}_+$) is *not* positive-mass. On the toy, inverting only the Arch atom produces a J -kernel with $\text{ind}_- = 2$ whose scalar Pick is not a multiple of D_0 (relative mismatch 0.777). A source-pure rule that inverts one prime still fails the DCCXCVII stop rule on index simplicity (lag-local fold(c_p^P) has $\text{ind}_- = 18, 19, 15$).

2 Question 2: the numbers

Proposition 1 (independent recomputation). *On the r409 five-atom toy, with local constructors (no import of `source_potapov_probe`):*

- (i) $m_X(7/3) = -7471/22950$ exactly, as both the unipotent product and the Cauchy sum.
- (ii) At interpolating depth $d_0 = |Y|$, $D_0 = W_Y - S_0$ has inertia $(1, 1, 0)$ and $(D_0)_{22} = -1744/2025$; the Cauchy Pick P_X has inertia $(2, 0, 0)$.
- (iii) Flagship MAIN $k_z = 9$: $\lambda_{\min}(P_X) = 8.67731$, $\text{ind}_-(I - \mathfrak{T}_0^* \mathfrak{T}_0) = 1$. Source-order growing- Y Uvarov: 0/104 negative pivots, minimum $3.248 \cdot 10^{-7}$. Position order (r409): one negative pivot at index 36. Lag-local interpolants: $\text{ind}_- = 18, 19, 15$ for $p = 2, 3, 5$.
- (iv) The Homma Redheffer star of two unipotent factors is the mass sum (residual 0). The star of the Blaschke–Potapov factors equals their chain product (residual $6.0 \cdot 10^{-17}$ at $z = i$), and the J -kernel Pick of that product is positive definite ($\lambda_{\min} = 4.679 \cdot 10^{-3}$ on the toy; prefix test on w_9 likewise $\text{ind}_- = 0$).

3 Question 3: Cauchy– π and a second window

Proposition 2 (Cauchy– π and the residual). *At kernel dimension 0,*

$$S_0 = \Delta^{-1} C \Pi W_X \Pi C^T \Delta^{-1}$$

with $C_{ia} = 1/(x_a - y_i)$, $\Pi = \text{diag}(\pi_Y(x_a))$, $\Delta = \text{diag}(\pi'_Y(y_j))$. The identity equals the Lagrange Gram, exactly over $\mathbb{Q}$, on the five-atom toy and on the independent six-node window $x \in \{-3/2, -1, -1/3, 1/5, 2/3, 3/2\}$ with $X = \{0, 2, 5\}$, $Y = \{1, 3, 4\}$. On that second window D_0 has inertia $(1, 2, 0)$ while P_X has inertia $(3, 0, 0)$: every negative square is the interpolant residual $W_Y - S_0$, not an Euler object. The Blaschke–Potapov J -kernel on the same Y -nodes remains positive definite.

4 Verdict

Ausgang CONFIRMED. SOURCE_FACTOR_NOT_FOUND stands, including the non-commutative check. R432 stays blocked (contract: FACTOR_EXACT). No second discovery round is opened by this audit. P1 stays $\lambda_2(C) \geq 1$ on the dual quotient. No RH claim.

Q	Verdict	Evidence
1	CONFIRMED	Lemma 1; non-commuting BP Pick $\text{ind}_- = 0$; genuine star = chain = mass sum on unipotents.
2	CONFIRMED	All R431 pins reproduced (Prop. 1).
3	CONFIRMED	Cauchy- π exact on two $\mathbb{Q}$ -windows; negatives live in D_0 , never in P_X .